STAT/MA 41600

In-Class Problem Set #36: November 11, 2026

1. While testing a new beverage for its sugar content, the amount of sugar (per can of soda) has a Normal distribution with mean 39 grams and standard deviation 2 grams.

1a. If a person purchases 5 cans of soda, and the amount of sugar is independent between the cans of soda, what is the probability that the 5 cans of soda have more than 197 grams of sugar altogether?

1b. In a container of 12 cans of soda, if the amount of sugar is independent between the cans of soda, what is the probability that the total amount of sugar is between 460 and 470 grams of sugar altogether?

2. A veterinarian considers cats to be fat if they weigh more than 15 pounds. Suppose that the weights of cats are considered to be independent and Normally distributed, each with average weight of 12 pounds and standard deviation of 5 pounds.

2a. If 10 people bring their cats to the veterinarian on a given day, what is the probability that they weigh more than 125 pounds altogether?

2b. What is the probability that the 10 cats in part **2a** weigh between 112 and 128 pounds?

2c. If the total weight of the cats is 145 pounds, does it follow that there is at least one fat cat in the group of 10 cats? Why or why not?

3. Suppose that 1000 grams of sugar weighs 2.2 pounds. Suppose that a person buys 150 cans of soda. What is the probability that the sugar content in the 150 cans of soda (using the assumptions from Question 1) weighs more than a cat (using the assumptions from Question 2)?

(Of course, you may also assume that the weight of the cat is independent from the weight of the sodas! The cat is not drinking soda!)

4. Suppose that the heights of blades of grass are Normally distributed, with each height having expected value 4 inches and standard deviation 0.75 inches.

Suppose that the heights of the blades of grass are independent. Suppose that 1075 blades of grass are lined up, end-to-end. What is the probability that they are longer than a football field?

(Note that a football field is 120 yards long, and note that there are 36 inches per yard.)

5. Assume that the heights of college females are Normally distributed, with expected height of 64 inches and standard deviation of 4.8 inches.

What is the probability that the student will be taller than 15 blades of grass (with independent heights) that are lined up, end-to-end, with the assumptions from Question 4?

Again, it is safe to assume that the height of the student is independent of the heights of the blades of grass.